

Muhammad Zia ul Haq

Physician-Epidemiologist · MD, MPH

Ann Arbor, Michigan | muhammad.zia@fulbrightmail.org | +1 734 590 1762 | LinkedIn: muhammad-zia-ul-haq

Physician and epidemiologist working across real-world evidence, causal inference, and AI in clinical workflows. I scope what a dataset can answer, design the study so the estimate survives scrutiny, and translate results for clinical and regulatory audiences. Applied experience with EHR, claims, and revenue-cycle data, clinical AI evaluation, and NIH and BARDA grant science.

EDUCATION

University of Michigan

Aug 2024 – Present

PhD Candidate, Epidemiology · GPA 4.0 · Rackham Merit Fellow

Dissertation: longitudinal recovery after intracerebral hemorrhage (BASIC study). Methods: marginal structural models, inverse probability weighting, mixed-effects survival analysis.

Emory University, Rollins School of Public Health

Dec 2021

Master of Public Health, Epidemiology · GPA 4.0 · Certificate in Genetic & Molecular Epidemiology

Nishtar Medical University

Feb 2017

MBBS (Doctor of Medicine) · Graduated top quartile

EXPERIENCE

Research Strategy Advisor · Switchboard, MD LLC

Jul 2022 – Present

- Build R and Python pipelines to clean, categorize, and visualize large EHR and revenue-cycle datasets, surfacing patterns in clinical messaging and medication use for quality improvement.
- Contribute epidemiologic design and methods writing for NIH and BARDA proposals on integrating AI and real-world data into clinical workflows.

Consultant, NCD Epidemiology & Policy · World Health Organization (WPRO & EMRO)

Jan 2023 – Oct 2024

- Synthesized IHME GBD, WHO, and national survey data into NCD country profiles for member states (Yemen, Tonga, Fiji), informing policy frameworks.
- Designed multinational survey analyses addressing confounding and bias; authored technical briefs on NCD management in emergency and fragile settings.

Senior Instructor · Aga Khan University, Community Health Sciences

Sep 2023 – Jul 2024

- Led curriculum and teaching for graduate courses in protocol development, critical appraisal, and data management.
- Co-led acquisition of two NIH research grants; mentored four master's students and two clinical residents.

Graduate Research Associate · Morningside Center for Innovative & Affordable Medicine

Jul 2020 – Dec 2021

- Core team member on a curated database of FDA-approved drugs for cancer repurposing; synthesized evidence from 700+ studies and authored 300+ article summaries.
- Independently built REDCap instruments for structured data annotation and quality control.

Investigator · Emory Global Health Initiative

Jun 2018 – Jul 2019

Supervised a prehospital trauma-triage smartphone app; established a trauma referral network and registry with Pakistani and U.S. stakeholders.

Staff Physician (House Officer) · Nishtar Hospital

May 2017 – May 2018

Managed inpatient and outpatient care across medical and surgical units; supported emergency triage and critical-care rounds.

SELECTED PUBLICATIONS

NEJM AI 2025 · Anderson BJ, **Zia ul Haq M**, Zhu Y, et al. Development and Evaluation of a Model to Manage Patient Portal Messages. NEJM AI. 2025;2(3). doi:10.1056/Aloa2400354

Frontiers in AI 2024 · **Zia ul Haq M**, Hornback A, Harzand A, et al. Graph-theoretic visualization of patient and health-worker messaging in the EHR. Front Artif Intell. 2024;7:1422208.

Circulation 2025 · **Zia ul Haq M**, Kwicklis M, Zahuranec DB, Morgenstern LB, Lisabeth LD. Longitudinal recovery after intracerebral hemorrhage: BASIC study. Circulation. 2025;151(Suppl 1):P3177.

Cochrane 2025 · **Zia ul Haq M**, Mansoor J, Lima Dos Santos CC, et al. Melatonin for blood pressure control in adults (Protocol). Cochrane Database Syst Rev. In press.

BMJ Open 2024 · Mansoor J, **Zia ul Haq M**, Ahsan Z, et al. HEPA filters in children with asthma: systematic review protocol of RCTs. BMJ Open. 2024;14(12):e087493.

Cureus 2023 · Lima Dos Santos CC, Matharoo AS, **Zia ul Haq M**. Influence of sex, age, and race on coronary artery disease: a narrative review. Cureus. 2023;15(10):e47799.

SKILLS & METHODS

Programming: R (tidyverse, nlme), Python (pandas, statsmodels), SAS 9.4, STATA, REDCap, Neo4j, Gephi	Regulatory: GCP, CITI Human Subjects, FDA real-world evidence, IRB submissions, data safety monitoring, trial registration
Data: EHR, claims, revenue-cycle, registry / real-world data	Certifications: Good Clinical Practice; CITI Human Subjects Research; Certificate in Genetic & Molecular Epidemiology; Ross Leaders Academy
Methods: cohort & case-control design, RCT, pharmacovigilance, marginal structural models, IPW, g-methods, survival analysis, mixed-effects models, network analysis	Languages: English (fluent), Urdu/Hindi (native), Punjabi & Saraiki (fluent), Spanish (conversational), Arabic (basic)

HONORS & AWARDS

Rackham Merit Fellow (U-M) · Phi Kappa Phi Honor Society, Centennial Class (top 10% of graduate students) · AHA Genomic & Precision Medicine Early Career Engagement Award · Fulbright Scholarship · Ross Leaders Academy 2025-26 · REAL Financial Award (Emory)

Conference presentations: EPI Lifestyle 2025, CSTE 2025, SER 2025, Emory Rollins 2021.